PREMIER UNIVERSITY, CHITTAGONG
DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

PROJECT REPORT

COURSE:
CSE 4345 - Big Data Analytics

PROJECT NAME:
EpiScope: Big-Data Surveillance, Forecasting and Response Planning

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Abstract

EpiScope is a prototype data-to-decision platform for district-level surveillance and response planning for dengue, diarrhoeal disease, influenza-like illness (ILI) and malaria in Bangladesh. It combines a deterministic synthetic encounter generator, a Spark-based Bronze-Silver-Gold processing pipeline, quality and privacy controls, outbreak detection, short-term forecasting, resource-demand conversion, linear-programming allocation, a seasonal preparedness calendar and a D3.js dashboard. A full-scale SF=1 execution generated 25,452,837 synthetic encounter rows. The recorded Silver and Gold pipeline runtime was 2,850.6 seconds (47.51 minutes), exceeding the assignment target of 45 minutes by 2.51 minutes. The subsequent analytics workflow completed in 147.8 seconds. The selected full-scale EARS C2 operating point recorded sensitivity of 0.75 and a median detection delay of 15 days; its false-alert log was 7.27 per day, which is not the same unit as the assignment metric of false alerts per district-year. Full-scale backtesting for Dengue and diarrhoeal disease produced lower MAE than the recorded seasonal-naive baseline at all four horizons, while 80% interval coverage ranged from 0.7589 to 0.8633. Allocation and Spark optimisation comparisons retained in the repository are from a separate SF=0.01 sandbox experiment and are labelled accordingly. The report distinguishes implemented features from requirements that remain unverified or incomplete.

Keywords

Big data analytics, Apache Spark, epidemiological surveillance, outbreak detection, forecasting, linear programming, privacy-preserving dashboard, Bangladesh.

I. INTRODUCTION AND PROBLEM ANALYSIS

The assignment describes a hypothetical Ministry of Health need for a national platform that turns routine health, climate and logistics data into early warnings and response plans. The platform is expected to detect unusual rises in four disease groups, forecast cases and hospital demand one to four weeks ahead, allocate limited supplies and rapid-response capacity, and prepare a seasonal calendar. It must serve both national and district decision-makers, work with incomplete and late data, and avoid exposing identifiable patient information [1]. The engineering problem is not simply to train a prediction model. It is to integrate heterogeneous data whose formats, completeness, time of arrival and geographic granularity differ. Duplicate encounters, late back-fills and reporting changes can create false signals. Forecasts must be translated into operational quantities, and recommendations must balance total unmet demand against minimum allocation equity. EpiScope addresses this by treating the pipeline, analytics and dashboard as one connected system.

A. Scope and objectives

- Diseases: Dengue (A90/A91), diarrhoeal disease (A00/A09), ILI (J10/J11) and malaria (B50-B54), plus background records.
- Spatial and time grain: 64 districts and epidemiological weeks; forecasting horizon: one to four weeks.
- Outputs: outbreak scores and alerts, district-level forecasts, demand estimates, allocation scenarios and a seasonal preparedness calendar.
- Data status: encounters, facility activity and stock snapshots are synthetic. Boundary geometry is provided through a 64-feature ADM2 TopoJSON file. Bundled population values are approximate placeholders rather than verified official census denominators. Weather is observed only when the Open-Meteo cache is successfully populated; otherwise synthetic fallback weather is used.

B. Stakeholders and needs

Stakeholder	Need / risk addressed
Ministry policymakers	National situation and fair, explainable supply decisions.
Surveillance epidemiologists	Reliable warnings and protection against reporting errors.
District health officers	Local warnings, forecasts and stock information.
Hospital managers	Bed and medical-supply demand estimates.
Vector-control units	Useful hotspot maps and seasonal timing.
Patients and public	Timely response without exposing personal health information.
Privacy regulator	Minimum data exposure and protection of small counts.

II. CEP ATTRIBUTE ANALYSIS AND CONFLICT RESOLUTION

The solution addresses the seven complex-problem attributes through explicit design decisions rather than by listing the attributes alone.

Attribute	How EpiScope addresses it
P1 - Depth of knowledge	Combines Spark-based distributed processing, Parquet data layers, time-series epidemiology, Poisson/negative-binomial forecasting, linear programming, D3 visualisation and health-data privacy controls.
P2 - Conflicting requirements	Treats sensitivity vs false alerts, map granularity vs privacy, efficiency vs equity, freshness vs cost and accuracy vs interpretability as explicit trade-offs.
P3 - Depth of analysis	Uses threshold selection, four forecast horizons, rolling-origin backtesting, seasonal-naive comparison and a budget/equity scenario grid. Some comparisons are only available from the low-scale sandbox and are labelled.
P4 - Unfamiliar issues	Models late records, duplicates, reporting completeness, back-fill artefacts and district-level skew. Artefact-suspect periods are gated from detection and forecasting features.
P5 - Beyond standard codes	Combines public-health coding assumptions, privacy suppression, operational conversion factors and an LP allocation model. Conversion factors are documented as assumptions requiring health-authority validation.
P6 - Stakeholders	Separates national and district dashboard data bundles and presents maps, forecast series, seasonal patterns, allocation summaries and ranked warnings for different decisions.
P7 - Interdependence	Data quality affects detection and forecasts; forecasts and current stock feed the optimiser; serving suppression and dashboard aggregation determine what decision-makers can see.

The engineering activities also reflect A1-A5: the team worked within a two-week limit using student-level compute and open data (A1); generator, processing, analytics and dashboard components had to integrate (A2); climate, health and logistics signals were combined in a single decision workflow (A3); misleading alerts or unfair supply recommendations could affect public trust and service readiness (A4); and the project applied course tools to a public-health domain beyond routine programming exercises (A5).

A. Resolution of conflicts C1-C5

Conflict	Design decision and evidence / remaining gap
C1: Sensitivity vs specificity	EARS C2 threshold was selected at tau=8.0 in the full-scale run. Recorded sensitivity was 0.75 and median delay 15 days. The log reports 7.27 false alerts/day, not false alerts per district-year; the required metric needs a consistent recalculation before final comparison. A gated path recorded 0 alerts in the artefact window versus 2 for the naive path.

Conflict	Design decision and evidence / remaining gap
C2: Granularity vs privacy	Published counts from 1-4 are suppressed and visually hatched. District-level views and separate district bundles reduce exposure. Complementary suppression/subtraction resistance and a formal privacy audit are not demonstrated; therefore the policy should avoid publishing multiple overlapping totals that reveal a hidden cell.
C3: Efficiency vs equity	The LP supports population-share equity floor alpha and five budget levels. In the retained SF=0.01 sandbox, alpha=0.2 reduced allocation Gini from 0.9844 to 0.8516 at the 50% budget, while weighted unmet demand increased from 1.444m to 1.666m project units. This is a policy trade-off, not a full-scale result.
C4: Freshness vs cost	The implemented pipeline supports batch/incremental processing, rather than a demonstrated five-minute streaming path. This keeps the student-laptop design manageable; Kafka/Structured Streaming refresh is not implemented. Daily batch is the proposed operating cadence, subject to actual service-level needs.
C5: Accuracy vs interpretability	District-disease Poisson GLMs use interpretable seasonal harmonics, autoregressive case terms and lagged weather. This is easier to explain than a black-box deep model. Model MAE beat the recorded seasonal-naive baseline in the evaluated full-scale horizons, but interval calibration and external validation remain necessary.

III. ARCHITECTURE AND DESIGN DECISIONS

EpiScope follows a medallion-style data architecture. Synthetic encounter, facility and stock data and public reference inputs enter Bronze with source and ingestion metadata. Silver applies schema casting, validation, de-identification, deduplication and quarantine. Gold creates analytical marts for report-week surveillance, onset-week epidemiology, stock and data-quality summaries. The analytics layer runs detection, forecasting, demand conversion, allocation and calendar generation. The serving layer aggregates outputs, applies small-cell suppression and writes JSON bundles consumed by the D3.js dashboard.

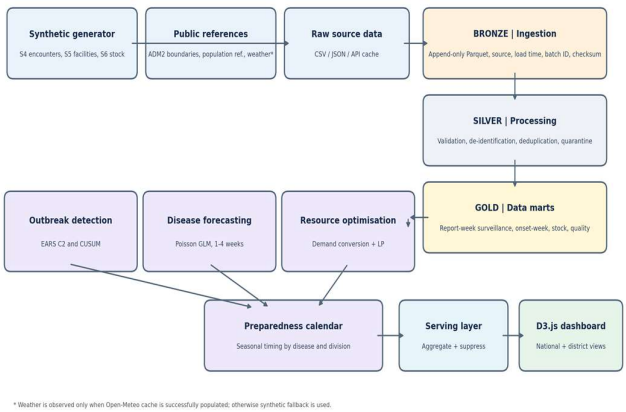


Fig. 1. EpiScope architecture. Weather is observed only when fetched and cached successfully; otherwise the generator uses synthetic weather.

A. Key design choices

- Apache Spark / PySpark is used for large transformations; Pandas is limited to small/reference or grouped model-fitting tasks. This follows the assignment restriction on heavy processing [2].
- Parquet is used for Silver and Gold. Delta Lake/Iceberg was not adopted because the current project uses a local Parquet workflow; no ACID time-travel implementation is claimed.
- Bronze is append-oriented and records source, load time, batch ID and checksum. Silver/Gold are partitioned by epidemiological year and division; Bronze uses load/snapshot date partitions.
- Two Gold perspectives are retained: report-week surveillance (what was available at that time, suitable for backtesting) and onset-week epidemiology (revised as late records arrive, suitable for seasonal analysis).
- Serving exports aggregated JSON rather than line-level encounter rows. National and district bundles are separate; the dashboard reads local static files.

B. Alternatives considered

A relational-only warehouse would make schema management familiar but would not satisfy the distributed heavy-processing requirement. A streaming-first design could reduce alert latency, but adds operational complexity and was not required to demonstrate the current batch/incremental prototype. A deep forecasting model could increase complexity without a clear interpretability benefit for a small student project. Parquet plus Spark local mode was therefore selected as the practical prototype architecture. HDFS/MinIO, streaming and Delta/Iceberg remain extension paths rather than completed features.

C. Project Team Roles and Responsibilities

The team uses one overall project lead, a deputy lead for analytics, and two technical module leads. The order below identifies project-level coordination and ownership, not a percentage contribution score. All members are expected to understand the integrated system and participate in cross-module testing. Appendix B must still be completed using actual work and commit evidence.

Member and order	Assigned role and primary responsibilities
1. Arnab Barua 0222220005101091	PROJECT LEAD & LEAD SYSTEMS ARCHITECT. Leads overall project direction, architecture and task coordination; oversees integration across the Spark pipeline, analytics and dashboard; coordinates final testing and release readiness.
2. Mohammed Masrafi 0222220005101080	DEPUTY PROJECT LEAD & ANALYTICS LEAD. Leads outbreak-detection and forecasting work, backtesting and metric interpretation, demand conversion, resource-allocation optimisation and the seasonal preparedness calendar.
3. Mohammed Iftekhar Sami 0222220005101090	DASHBOARD & VISUALISATION LEAD. Leads the D3.js dashboard, national and district views, map and time-series presentation, seasonal and allocation views, linked interactions and dashboard usability checks.
4. Foisal Abideen 0222220005101072	DATA QUALITY, PRIVACY & VALIDATION LEAD. Leads validation-rule and quarantine review, deduplication and lineage checks, privacy-control verification, automated-test evidence and data-source documentation.

Leadership order: Arnab Barua is the overall project lead, followed by Mohammed Masrafi as deputy project lead. Iftekhar Sami and Foisal Abideen lead their respective technical modules. The signed contribution matrix remains the record of actual participation and agreed contribution shares.

IV. DATA GENERATION, QUALITY, LINEAGE AND PRIVACY

A. Data sources and generator

The assignment defines S1-S7 as boundaries, population, daily weather, synthetic health encounters, synthetic facility registry, synthetic stock/bed snapshots and optional OpenDengue reference. The project bundle contains a 64-feature Bangladesh ADM2 TopoJSON and a district reference CSV. The project source notes that bundled population values are approximate placeholders; they must not be described as official BBS observations. The Open-Meteo fetcher can cache historical weather, but synthetic fallback is used when no cache is available. OpenDengue is optional and is not used in the default workflow.

The deterministic generator is configured by scale factor and random seed. The recorded SF=1 execution generated 25,452,837 encounter rows for 2021-2025 across the 64-district setting. The generator also creates facility and stock/bed snapshot records and injected outbreak/reporting-artefact conditions. Encounter data is synthetic and no real patient-level data is used.

B. Bronze-Silver-Gold and quality checks

Bronze retains source payloads and adds source, load timestamp/date, batch identifier and file checksum. Silver converts types and validates required identifiers, dates, ages, sex, ICD-10 format, facility references and geographic references. Invalid records are routed to quarantine with rule identifiers. The rule catalogue includes 12 row-level rules (Q1-Q10, Q13-Q14) and four set-level checks (Q11, Q12, Q15, Q16), covering encounter duplicates, cross-facility patient deduplication, reporting completeness and batch-volume drift.

For the recorded SF=1 run, pipeline logs reported 24,953,759 Silver rows, 569,314 quarantined rows, 499,078 encounter resubmissions removed, 633,305 patient duplicates merged, 2,094 late-extreme records excluded from the relevant analysis path and 83,840 Gold onset rows. These are counters from different processing stages and should not be added as if they were mutually exclusive partitions. The retained counters document substantial data cleaning and aggregation.

C. Lineage and incremental processing

Bronze ingestion uses file path and SHA-256 checksum to avoid loading an already-seen file. Silver processes unseen batch identifiers and anti-joins encounter identifiers to prevent resubmission duplication. Gold marts are rebuilt from Silver so late-arriving records can be reflected in affected analytical weeks. Core outputs retain batch/lineage metadata. The code includes idempotency logic and automated tests; however, a complete instructor-style hidden 14-day drop, repeated twice with before/after evidence, has not been recorded in the supplied execution evidence.

D. Privacy controls

- Patient identifiers are salted-hashed in Silver using SHA-256, truncated to 128 bits; the salt is loaded from environment/local secret storage and must not be committed.
- Exact age and direct patient identifiers are not intended to remain in Silver/dashboard outputs; age is converted to bands.
- Counts from 1 to 4 are suppressed in published patient-count outputs. The map uses hatching to distinguish suppressed cells from zero values.
- The serving layer emits aggregated JSON and separate district bundles. The bundles are not equivalent to server-side authorisation; server-side role enforcement is not implemented.

Small-cell suppression is a practical control, but a formal complementary-suppression audit, record-level k-anonymity test and differential-privacy mechanism are not demonstrated. These controls must not be described as complete privacy certification [8].

V. ANALYTICS METHODS

A. Outbreak detection

The Spark detection module implements EARS C2 using a rolling baseline window over weeks t-9 to t-3, leaving a guard band before the current week. The standardised score is computed as (observed-baseline mean)/max(baseline standard deviation, 1). Weeks with fewer than five baseline observations do not receive a score. A gated variant excludes artefact-suspect weeks from baseline estimation and alerting. A CUSUM implementation accumulates positive deviations and is available for comparison [3], [4].

The full-scale run selected the C2 operating point tau=8.0. The recorded metrics were sensitivity=0.75, false alerts/day=7.27 and median detection delay=15 days. The same run recorded two alerts for the naive detector and zero for the gated detector in the identified reporting-artefact window. The false-alert unit differs from the assignment target of false alerts per district-year; it is retained exactly as reported rather than silently converted.

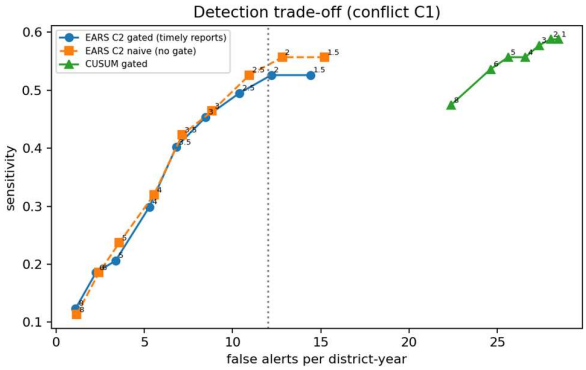


Fig. 2. Exploratory detection threshold trade-off from the retained SF=0.01 sandbox experiment. This is not the SF=1 operating-point curve.

B. Disease forecasting

The forecasting module fits a district-disease Poisson GLM in parallel with Spark grouped Pandas execution. Predictors include annual and semi-annual seasonal harmonics, current and lagged reported cases, lagged precipitation and temperature, population offset and reporting-completeness adjustment. A moment-based over-dispersion estimate is used to form negative-binomial prediction quantiles. Forecast horizons are one, two, three and four weeks. Rolling-origin backtesting covers the final 52 weeks and refits every four weeks, comparing against a seasonal-naive value from the corresponding week in the previous year [5].

Disease	Horizon	Model MAE	Seasonal-naive MAE	80% coverage
Dengue	1 week	39.0211	93.4467	N/R
Dengue	2 weeks	54.5017	93.4467	N/R
Dengue	3 weeks	58.2504	93.4467	N/R
Dengue	4 weeks	59.0791	93.4467	N/R
Diarrhoeal	1 week	39.0633	91.2957	N/R
Diarrhoeal	2 weeks	53.3723	91.2957	N/R
Diarrhoeal	3 weeks	51.5538	91.2957	N/R
Diarrhoeal	4 weeks	69.7577	91.2957	N/R

The model MAE is lower than the recorded seasonal-naive MAE at all eight evaluated disease-horizon combinations. The retained summary reports an overall 80% prediction-interval coverage range of 0.7589-0.8633, but the individual coverage value for each disease-horizon pair was not retained in the available evidence; N/R means not reported at that granularity. Interval calibration is therefore not uniformly confirmed at the nominal 0.80 level and must be interpreted alongside MAE.

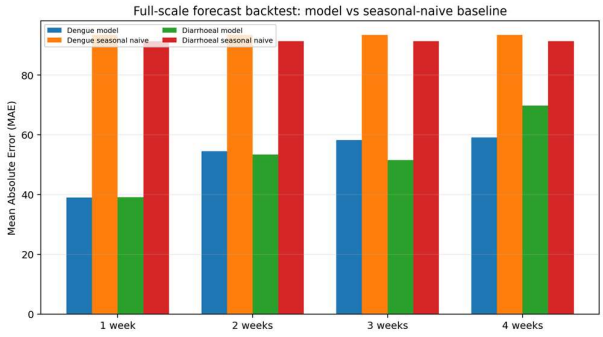


Fig. 3. Full-scale SF=1 recorded MAE comparison for Dengue and diarrhoeal disease.

C. Demand conversion and allocation optimisation

Four-week forecast quantiles are converted into estimated resource demand. The current code uses q90 forecasts and explicit, configurable assumptions: admission rates of 0.22 (Dengue), 0.10 (diarrhoeal), 0.05 (ILI) and 0.15 (malaria); lengths of stay of 4, 2, 3 and 3 days, respectively; and disease-specific IV saline, NS1, ORS and antimalarial conversion factors. These rates are engineering assumptions, not clinically validated Ministry norms.

The LP minimises the sum of district severity weights multiplied by unmet demand. Constraints represent the available budget, a population-proportional equity floor alpha, and division-level transport capacity. PuLP/CBC solves the optimisation. The dashboard scenario grid varies five budget levels (10%, 25%, 50%, 75%, 100% of the estimated gap cost) and five alpha levels (0, 0.1, 0.2, 0.3, 0.5). Alpha is a policy lever: higher alpha guarantees more population-proportional minimum allocation but can increase weighted unmet demand.

D. Seasonal preparedness calendar

The calendar module derives seasonal timing from the epidemiological series and produces recommended pre-positioning and campaign windows by division and disease group. It is a planning aid rather than a deterministic procurement schedule; it inherits the limitations of synthetic case histories and approximate population references.

VI. EVALUATION AND PERFORMANCE

A. Full-scale pipeline and analytics run

Measure	Recorded result
Execution environment	Windows 11; 16 GB RAM; local Spark
Runtime stack	Python 3.11.9; JDK 17.0.12; PySpark 3.5.1
Scale factor / seed	SF=1.0 / seed 42
Generated encounters	25,452,837
Silver rows	24,953,759
Quarantined rows	569,314
Resubmissions removed	499,078
Patient duplicates merged	633,305
Late-extreme records excluded	2,094
Gold onset rows	83,840
Silver + Gold runtime	2,850.6 s (47.51 min)
Assignment runtime target	45 min or less; not achieved
Analytics runtime	147.8 s total

Silver processing took 2,397.3 seconds and Gold processing 453.3 seconds, for a combined 2,850.6 seconds. This exceeded the 45-minute target by 150.6 seconds (2.51 minutes), approximately 5.6% over the target. The analytics stages completed in 147.8 seconds: detection 30.3 s, backtesting 87.6 s, production forecasting 19.7 s and optimisation 10.2 s. The serving export was approximately 7.21 MB, below the 10 MB browser-data target.

B. Allocation comparison (retained low-scale evidence)

The saved allocation_summary.csv comes from a separate SF=0.01, one-core build sandbox, not the full SF=1 run. At alpha=0, the LP weighted unmet-demand objective was lower than the population-proportional baseline at each of the five tested budget levels. The table reports the saved project units and percentage reduction.

Budget	LP weighted unmet	Population baseline	LP reduction
10%	2,423,880	2,664,880	9.0%
25%	2,056,451	2,658,951	22.7%
50%	1,444,069	2,649,068	45.5%
75%	831,687	2,639,186	68.5%
100%	427,645	2,629,303	83.7%

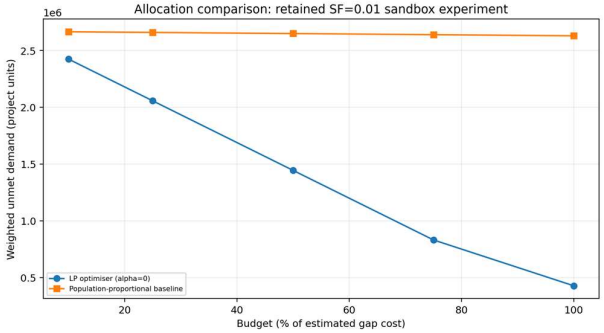


Fig. 4. Saved SF=0.01 sandbox allocation comparison (alpha=0); not a full-scale allocation measurement.

The alpha trade-off is visible at the 50% budget: at alpha=0, weighted unmet demand was 1,444,069 and allocation Gini was 0.9844; at alpha=0.2, weighted unmet demand increased to 1,665,884 while allocation Gini declined to 0.8516. This illustrates the cost of imposing a stronger equity floor. The current full-scale optimisation stage completed, but a separate SF=1 table of LP-vs-baseline results was not available in the supplied evidence.

C. Spark optimisation experiments

The retained Spark optimisation CSV is also from SF=0.01 on a one-core sandbox. It is useful as exploratory evidence but does not meet the full performance-evidence standard by itself. The best measured reductions were partition pruning (0.439 s to 0.173 s, 60.6%) and file compaction (1.368 s to 0.440 s, 67.8%). Broadcast join reduced the recorded join time by 32.2%; AQE coalesce reduced it by 16.0%. Two-stage salting was slower by approximately 4.4% in that test and is not claimed as an optimisation win. File compaction reduced the file count from 563 to 56 and output size from 17.53 MB to 12.26 MB.

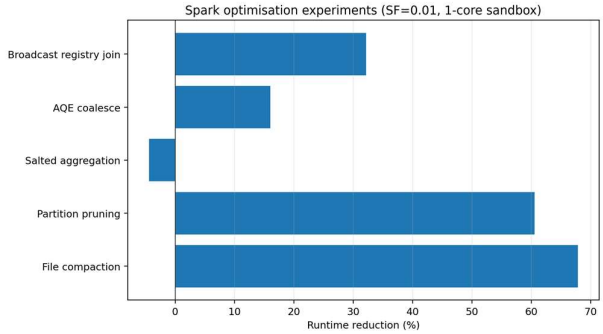


Fig. 5. Relative runtime changes in the retained SF=0.01 Spark experiment; salting was slower.

Spark UI screenshots and before/after traces at SF=1 were not included in the supplied evidence. A complete SF=0.25, 0.5 and 1.0 timing series was also not available. Therefore the project demonstrates a full-scale run, but the required full scalability study and Spark UI evidence remain incomplete.

D. Automated tests

The recorded test run completed with 16 passed and 18 warnings in 950.24 seconds. The warnings were Pandas deprecation warnings. The suite covers generator behaviour, quality rules, idempotency, privacy/isolation, suppression, detection logic, optimiser constraints and payload checks. This result applies to the code state and test command used during that run; it is not proof that an instructor-provided hidden 14-day drop has been replayed twice.

VII. DASHBOARD DESIGN AND USABILITY

The dashboard is implemented in D3.js v7 and contains five coordinated view modules: (1) a district incidence/alert map; (2) a weekly series with forecast and prediction bands; (3) a seasonal week-by-year heatmap; (4) a supply-allocation view with budget and equity controls; and (5) a ranked early-warning table. Disease selection and district/time-range interactions are wired through shared application state. National and district-officer modes are supported through separate JSON bundles. A compact data-quality indicator and data-freshness text appear in the header. This follows the D3 data-driven interaction model [7].

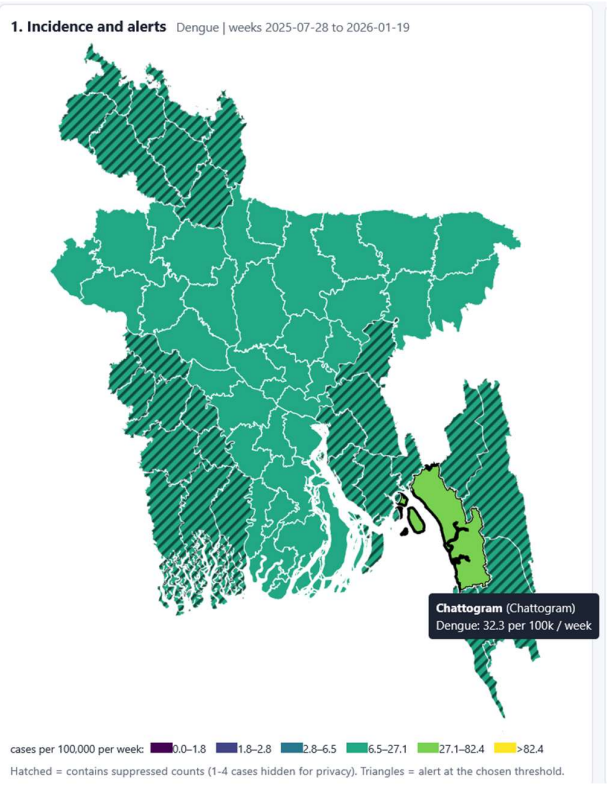


Fig. 6. Browser screenshot of the district incidence map. Hatching identifies cells containing suppressed counts (1-4); the displayed population-rate values inherit the approximate population-denominator limitation.

The dashboard export measured approximately 7.21 MB. The screenshot shows the map and its suppression hatch legend. A district-mode page was also opened during browser verification. However, the observed early-warning ranking table was blank in one run, and the efficiency-equity visual was degenerate (a single point at the origin) in the observed dashboard state. No documented usability study with at least two non-CS users was available. Therefore the dashboard is implemented and locally exercised, but ranking-table correctness, allocation-chart rendering, linked-view behaviour and non-technical-user usability should be rechecked before the final video.

VIII. ETHICAL, SOCIETAL AND SUSTAINABILITY CONSIDERATIONS

The platform is an educational prototype, not a clinical or public-health authority. All encounter-level patient data is synthetic; outputs must not be interpreted as real disease estimates. Population denominators are approximate placeholders and the weather series may be synthetic when the API cache is unavailable. The report therefore avoids presenting the map rates or allocation outputs as official Bangladesh surveillance statistics.

Privacy protection is central because health data can expose individuals and stigmatise communities. The system hashes identifiers, removes direct identifiers from analytical layers, reduces age precision and suppresses counts of 1-4. Separate dashboard bundles reduce accidental cross-district detail exposure, but do not replace server-side access control. Suppression must be reviewed for subtraction attacks before public release [8].

Equity matters because a purely efficiency-oriented optimiser may allocate nothing to low-demand districts. The alpha floor gives decision-makers a visible policy lever, but the increase in weighted unmet demand must be explained. False alerts can waste investigation time and reduce trust; missed alerts can delay response. Human review remains necessary. Sustainability is supported by local Spark execution, columnar storage, small aggregated browser payloads and open-source components, but the 47.51-minute Silver+Gold runtime exceeds the specified target and should be improved before claiming efficient national-scale operation.

IX. LIMITATIONS AND FUTURE WORK

- The SF=1 Silver+Gold runtime was 47.51 minutes, 2.51 minutes above the 45-minute requirement. Profile skew, partition sizing, file counts and join strategies at full scale.
- A comparable SF=0.25/0.5/1 stage timing series and Spark UI screenshots are not in the supplied evidence. Re-run performance experiments on the 16 GB target environment.
- Allocation and Spark optimisation comparison files are SF=0.01 sandbox results. Produce and retain full-scale outputs and a clearly stated allocation policy.
- Recompute false alerts per district-year and retain the complete SF=1 sensitivity/false-alert/delay threshold curve. The the current run log reports 7.27 false alerts/day, which is not the required district-year unit.
- Replay the instructor-style hidden 14-day drop without code changes; run it twice and compare records, alerts, allocation and dashboard outputs to demonstrate idempotency and late-data handling.
- Validate population denominators against BBS Census 2022 or WorldPop; document source, license, district matching and year. Verify whether the current weather cache contains real observations or synthetic fallback values.
- Add formal complementary-suppression/subtraction tests, k-anonymity discussion, differential-privacy design and server-side role authorisation before any real health data use.
- Fix the blank early-warning table and degenerate efficiency-equity plot, verify all five linked views, and conduct a documented usability check with at least two non-CS users.
- Validate demand-conversion assumptions with public-health professionals. Recalibrate forecast intervals and assess out-of-sample performance beyond synthetic data.
- HDFS/MinIO storage tests, Structured Streaming/Kafka, Delta/Iceberg time travel and SEIR scenario modelling are not demonstrated as completed features.

X. CONCLUSION

EpiScope demonstrates an end-to-end prototype that connects synthetic health data generation, Spark medallion processing, privacy-oriented transformation, outbreak detection, short-term forecasting, resource optimisation, seasonal planning and a D3.js dashboard. The full-scale generator met the 25-million-row requirement and the recorded analytics workflow completed. Forecast MAE beat the seasonal-naive baseline for the two evaluated disease groups and all four horizons. At the same time, the Silver+Gold runtime missed the 45-minute target, the false-alert unit needs harmonisation with the assignment, the allocation/Spark optimisation comparisons remain low-scale evidence, and several hidden-drop, usability and privacy checks need more proof. These limitations are explicitly retained so that the prototype can be assessed on measured evidence rather than unsupported claims.

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APPENDIX A. TWO-PAGE POLICY BRIEF FOR THE DIRECTOR

EpiScope: Using early warnings and forecasts to prepare district health services

To: Director, Disease Control and Preparedness Unit, Ministry of Health
Subject: A cautious pilot of district-level disease surveillance and supply planning

Purpose

EpiScope is a working educational prototype that brings synthetic health reports, facility and stock information, reference geography and weather inputs into a single district-level view. It is designed to help officials notice unusual changes, review likely near-term demand and discuss how limited supplies could be distributed. It is not an official surveillance service and it does not diagnose patients or replace a health officer's judgement.

What the current test shows

The full-scale test generated more than 25 million computer-generated health visits for 2021-2025. After processing, the system produced district-level monitoring and forecast results. In the recorded test, the selected warning setting identified about 75% of the simulated outbreaks, with a typical delay of 15 days. The system log also recorded 7.27 false warnings per day. That figure has not yet been recalculated in the annual, per-district format required for final evaluation. Officials should treat every warning as a reason to check local reports, not as automatic proof of an outbreak.

Forecast tests for Dengue and diarrhoeal disease had lower average error than a simple seasonal comparison in all four forecast horizons. However, the uncertainty ranges did not achieve exactly 80% coverage at every horizon. The project's allocation comparison and Spark tuning results were retained from a much smaller test run, so they should be treated as preliminary demonstrations rather than national-scale evidence.

Three recommendations

1. Use the warning list as a weekly review queue, not an automatic alarm.

Ask district surveillance teams to verify the highest-ranked signals against facility reporting, laboratory confirmation and local context. The simulated test detected outbreaks with a median delay of 15 days at the selected operating point, but false alerts remain a concern. A human review step reduces the risk of unnecessary response and protects trust in the system.

2. Test forecast-informed allocation with an explicit equity floor.

For a pilot, consider expected district needs, available stock and a minimum share for every district together. In the preliminary small trial, the computer-planned distribution left less estimated need unmet than a population-only split at each of five budget levels. Requiring a larger minimum share made the distribution fairer between districts, but also left more total need unmet. Officials should agree on this balance openly before using any recommendation.

3. Pre-position resources before the expected seasonal rise.

Use the seasonal calendar to start early discussions about purchasing, checking supplies and timing prevention campaigns. Confirm dates with district teams and current health information before spending money or moving supplies. The calendar is based on computer-generated history and must be checked against verified information before operational use.

How to read the dashboard

- Map: colours show the displayed level of reported cases; triangles mark warnings. Striped areas contain small counts hidden to protect privacy. They do not mean that there were no cases.
- Weekly chart: compare the reported counts with the forecast. The shaded range shows that future counts are uncertain, so do not rely on the central line alone.
- Seasonal view: compare the same weeks across years and use the calendar as an early reminder when planning campaigns.
- Supply plan: compare expected needs, current stock and the suggested distribution. Changing the budget or fairness setting changes only the computer scenario, not the amount physically held by a warehouse.
- Warning list: use the ranking to decide which districts to check first. This list must be checked again because it appeared blank during one browser test.

APPENDIX A (CONTINUED). CONFIDENCE, LIMITATIONS AND DATA PROTECTION

Confidence and limitations

Confidence is reasonable that the software workflow can run as demonstrated, but low for using its numbers to make real public-health decisions. The data-processing, analysis and dashboard steps were run in the project environment. Forecasts were closer to the test values than the simple seasonal comparison for the two diseases evaluated. However, all health visits are computer-generated, the forecast uncertainty ranges need improvement, and the population figures included with the project are approximate rather than verified official district totals. Weather values are observed only when the external weather service has been successfully downloaded; otherwise computer-generated weather is used. Map rates depend on the quality of the population figures.

The main data-processing run took 47.51 minutes, which is 2.51 minutes above the the 45-minute target specified in the assignment. The saved supply-planning and computer-speed comparisons came from a separate, much smaller test. The the instructor-provided hidden 14-day data test has not yet been demonstrated twice with before-and-after evidence. Some dashboard parts also need another check, especially the warning list and the chart comparing fairness with efficiency. EpiScope should therefore be treated as a prototype for further testing, not as a system ready to direct real supplies.

Data-protection assurance

The prototype uses computer-generated patient visits and does not need real patient records. Its design changes patient identifiers into protected codes, groups ages into ranges and hides published counts from 1 to 4. The dashboard receives summary information rather than individual patient visit records. The private value used to create protected codes, along with passwords and API keys, must stay outside the shared repository.

Before any real health information is used, the Ministry would need a formal privacy and security review, secure handling of passwords and private keys, checks to ensure users can see only the information they are authorised to see, activity records, and tests to make sure hidden small counts cannot be worked out from totals. Separate district files in the current prototype do not provide these protections by themselves. No real patient information should be uploaded to this prototype.

Decision requested

Approve further testing only. Before a trial in real health services, require (1) verified population and weather information, (2) a successful repeat test with late-arriving records and no duplicate results, (3) false-warning rates reported consistently for each district and year, together with full-scale supply-planning results, (4) fixes and user testing for every dashboard view, and (5) an independent privacy and security review. Any future trial must remain under human supervision and clearly show uncertainty and limitations beside each recommendation.

APPENDIX B. TEAM CONTRIBUTION MATRIX

All four members contributed to the project work under the responsibilities shown in Section III-C. To avoid parallel push and merge conflicts in the shared repository, GitHub integration and the final push were centralized through Arnab Barua. As a result, the visible GitHub commit/push history identifies the uploader and integrator; it should not be read as a complete measure of each member's technical work. The L/S entries below describe the team's reported work allocation. The proposed percentage shares must be reviewed and approved by all four members before signing. In the matrix, L means lead responsibility, S means support, and - means no primary assignment in that work area. The 35/25/22/18 split is a proposed role-weighted allocation, not a system-generated measurement; it becomes final only after all four members agree.

Work area	Member 1 Arnab Barua 0222220005101091	Member 2 Mohammed Masrafi 0222220005101080	Member 3 Mohammed Iftekhar Sami 0222220005101090	Member 4 Foisal Abideen 0222220005101072
Problem analysis, proposal and conflict analysis	**L**	S	S	S
Synthetic generator and public-data acquisition	**L**	S	-	S
Ingestion, lakehouse layers and incremental loads	**L**	S	-	S
Data quality, lineage and privacy controls	S	-	S	**L**
Outbreak detection and evaluation	S	**L**	-	S
Forecasting, backtesting and demand conversion	S	**L**	S	-
Optimisation model and preparedness calendar	S	**L**	S	-
Serving layer and D3.js dashboard	S	S	**L**	S
Scalability experiment and Spark optimisation	**L**	S	-	S
Report and policy brief	**L**	S	S	S
Video production	**L**	S	S	S
GitHub integration / push record	Single uploader: final integration and GitHub push.	No separate push; work included in consolidated upload.	No separate push; work included in consolidated upload.	No separate push; work included in consolidated upload.

APPENDIX C. INDIVIDUAL REFLECTIONS

The following are draft reflections prepared from the assigned team roles and recorded project evidence. Each member must read, correct and approve their own reflection so that it accurately represents their personal experience before signing or submitting the report.

Member 1: Arnab Barua | 0222220005101091

I worked as the Project Lead and Lead Systems Architect. My main responsibility was coordinating the overall workflow and bringing the generator, Spark pipeline, analytics outputs and dashboard into one working project. I also handled the consolidated GitHub integration and final push. We used one uploader to avoid simultaneous pushes and merge conflicts; this was a repository-management decision and does not mean the other members did not contribute. One major engineering challenge was running the full-scale Spark pipeline on a student development machine. I checked the stage timings and found that the Silver and Gold processing stages took 2,850.6 seconds, or 47.51 minutes, which exceeded the 45-minute target. This showed why a completed run should not automatically be described as meeting the performance requirement. We chose a local batch workflow rather than claiming an unimplemented streaming service. With more time, I would focus on partition and skew optimisation, repeat the clean-clone and hidden-drop tests, and document the results more fully.

Member 2: Mohammed Masrafi | 0222220005101080

I worked as the Deputy Project Lead and Analytics Lead. My main workstream covered outbreak detection, disease forecasting, demand estimation, resource optimisation and the seasonal preparedness calendar. The analytics results showed that model performance must be considered together with the operational cost of alerts. In the recorded detection run, the selected EARS C2 operating point used a threshold of 8.0, sensitivity of 0.75, 7.27 false alerts per day and a median detection delay of 15 days. The artefact-window comparison produced two alerts for the naive detector and none for the gated detector. For forecasting, the one-week Dengue MAE was 39.0211 compared with 93.4467 for the seasonal-naive baseline; the one-week Diarrhoeal MAE was 39.0633 compared with 91.2957. These are experimental results on synthetic data, not clinical validation. The main trade-off was earlier detection versus false-alert burden. With more time, I would recalibrate thresholds, report false alerts using the assignment's district-year definition, and validate uncertainty intervals and demand-conversion assumptions.

Member 3: Mohammed Iftekhar Sami | 0222220005101090

I worked as the Dashboard and Visualisation Lead. My assigned area was the D3.js interface, including the Bangladesh district map, disease time-series and forecast views, seasonal heatmap, resource-allocation controls and linked national and district views. The dashboard receives aggregated JSON from the serving layer rather than displaying encounter-level patient records. The recorded export was approximately 7.21 MB, which remained below the 10 MB browser-data target. An important design decision was to suppress small counts and use aggregated views so that the dashboard would be more useful without exposing identifiable patient information. This creates a trade-off because users receive less detail for very small counts, but privacy has to take priority. During browser checking, the early-warning ranking table appeared blank in one test, so it should be treated as a known issue rather than presented as fully verified. With another week, I would fix and retest the ranking view, improve mobile layout and accessibility, and conduct usability checks with non-technical users.

Member 4: Foisal Abideen | 0222220005101072

I worked as the Data Quality, Privacy and Validation Lead. My assigned workstream focused on checking source records, validating fields and references, separating invalid records into quarantine, reviewing duplicate handling and checking privacy protections before data reached the analytical and dashboard layers. In the full-scale pipeline run, 569,314 records were quarantined and 499,078 resubmissions were removed. The Gold processing report also recorded 633,305 patient duplicates merged; these are separate pipeline measures and should not be added together as if they were the same type of record. The automated test run recorded 16 passed tests, with 18 warnings. A key engineering trade-off was to reject or quarantine questionable records rather than allow poor-quality inputs to silently affect outbreak signals and forecasts. Privacy controls also require aggregated exports and suppression of counts from one to four. With more time, I would extend the quality report evidence, test complementary suppression against subtraction, and perform a more complete lineage and privacy audit.